Detecting Exoplanet Transits with the Keeble Telescope

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Introduction

Objective: To confirm the possibility of whether the Keeble telescope can detect an exoplanet transit. Alongside with the main goal, the camera's sensitivity and precision will be explored.

Brief Conceptual Overview: An exoplanet is a planet that resides outside of our solar system. These exoplanets have transits, which occur under the circumstances of the planet passing between the star and the observer. Many exoplanets have been discovered through the *transit method*; this method utilizes photometry by recording the changes in flux from the exoplanet's host star and then graphing the detected values into a light curve.

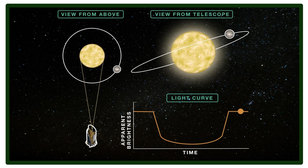


Figure 1: Example of exoplanet passing between host star and observer
Credit: viewspace.org

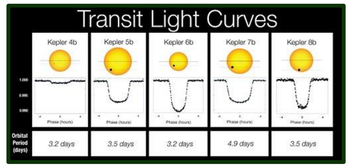


Figure 2: Example of light curves from various exoplanets
Credit: NASA

Observatory

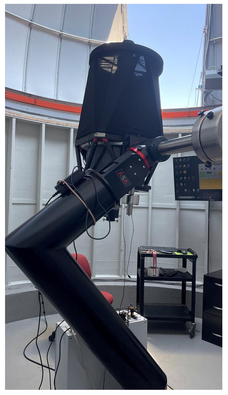


Figure 3: Telescope in Keeble Observatory

About Keeble: Although Keeble observatory's existence is over 60 years old, it was recently rebuilt in 2017. Inside features the telescope, an ASA 40 cm Ritchey-Chretien F/8.The dome is controlled by ACE SmartDome and ASCOM POTH hub, meanwhile the telescope relies on Autoslew and Cartes du Ciel for movements and ACC for shutter and focal adjustments.

About the CCD: The CCD, or Charge-Coupled Device, is the camera used to capture images of the night sky. It is a SBIG ST-10XMEI with thermoelectric cooling and a pixel resolution of 0.88 arcsec per pixel. The camera operates through Maxim DL. We rely on an infrared filter to minimize effects from light pollution. Therefore, the filter only allows infrared light to pass through.

Future

Future Work: Some considerations for future research are to investigate some of the common issues that interfered with project. The main offender was subpar telescope tracking, which ruined many of the images. Environmental factors, such as the summer heat, caused obstacles with the CCD's thermoelectric component. One of the possible solutions may be to purchase an autoguider. As for the data, it would be ideal to figure out how to de-trend the light curve. Another consideration is to apply to join exoplanet groups and to contribute findings from continuous or follow-up observations.

Data and Results

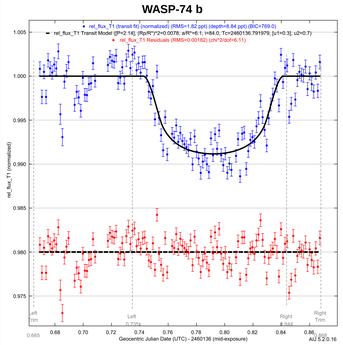


Figure 4: Light curve for the WASP-74 b transit

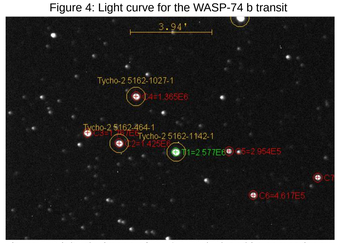


Figure 5: Night sky image of WASP-74 (T1) and its comparison stars (C2 to C7)

WASP-74 b

Host star:
- 9.8 magnitude
- 149.216 pc from Earth
- Temperature: 5806 K
- Mass: 1.044 $M_{\odot}$
- Radius: 1.59624 $R_{\odot}$

Exoplanet:
- Period: 2.14 days
- Semi-major axis: 0.03443±0.00036
- Mass: 0.72±0.12 M_{Jup}
- Radius: 1.36±0.10 R_{Jup}

Transit: Expected/ Experiment
- Date: 7/10-7/11/23
- Depth: 9.6 ppt / 8.84 ppt
- Duration: 2:17 hours
- Transit (Eastern Time): 01:48--02:56--04:05
- T_c (UTC): 06:56 / 07:00

Data: The total time of the observing run is 10.5 hours. All science image exposures are 120 seconds long at -7˚ C. 30 exposures were taken in the span of an hour, totaling to 132 images overall.

Results: As shown, the data is sufficient to produce a light curve. The expected depth is 9.6 ppt and the experimental data finds a depth of 8.84 ppt. Another valid comparison is the expected time of the transit center (T_c) versus the time given by the data. The expected time was 02:56 EST (06:56 UTC), meanwhile the calculated peak was 03:00 EST (07:00 UTC). Last, but not least, the data gave a planetary radius measurement of 1.3 R_{Jup}, whereas the expected value is 1.36±0.10 R_{Jup}.

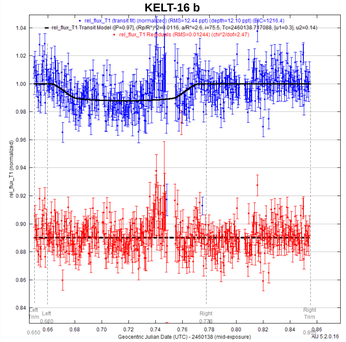


Figure 6: Light curve for the KELT-16 b transit

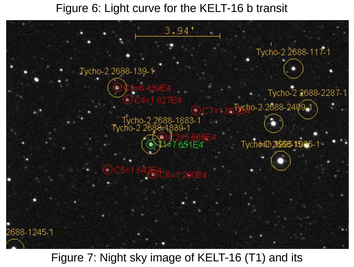


Figure 7: Night sky image of KELT-16 (T1) and its comparison stars (C2 to C7)

KELT-16 b

Host star:
- Period: 0.97 days
- Semi-major axis: 0.02044
- Mass: 1.415 M_{Jup}
- Radius: R_{Jup}

Exoplanet:
- 11.7 magnitude
- 444.6 pc from Earth
- Temperature: 6430 K
- Mass: 1.3 $M_{\odot}$
- Radius: $1.415^{+0.084}_{-0.067}$ $R_{\odot}$

Transit: Expected/ Experiment
- Date: 7/12-7/13/23
- Depth: 11.5 ppt / 12.10
- Duration: 2:29 hours
- Transit (Eastern Time): 00:00--01:15--2:30
- T_c (UTC): 05:15 / 05:12

Data: The total time of the observing run is 10.5 hours. All science image exposures are 30 seconds at -5˚ C. 120 exposures were taken in the span of an hour, totaling to 480 images. This observing run has shorter exposures to combat saturation and tracking issues.

Results: Some comparisons between expected and experimental values are the following. The data provided a transit depth of 12.10 ppt, deeper than the expected 11.5 ppt. As for the transit center, expected was 01:15 EST (05:15 UTC) while the calculated was 01:12 EST (05:12:36 UTC). AstroImageJ gave a planetary radius values of 1.48 R_{Jup}. This is comparable to the expected measurement of $1.415^{+0.084}_{-0.067}$ R_{Jup}.

Conclusion: It is confirmed that the Keeble telescope can detect an exoplanet transit. This was achievable from the improvements made with the hardware; additionally, it is worth mentioning that the characterization of the telescope contributed as well. Hopefully in future research, some of the obstacles can be further investigated.

Improvements

Improvements: Several modifications were made to the observatory. The first change was to the dome; it was limited to moving only through manual controls but is now automated. Another example is covering the telescope with a shroud to prevent scattered light from entering and interfering with imaging. One of the most noticeable improvements was collimating the telescope.

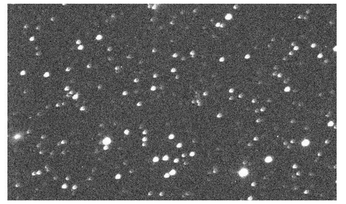


Figure 8: *HAT-P-37 b*, before collimation (*6/13-6/14/23*), 180 second exposure

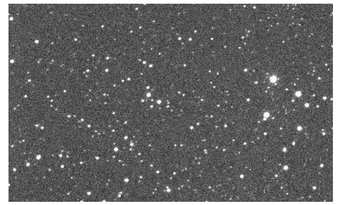


Figure 9: *KELT-16 b*, after collimation (*7/12-6/13/23*), 30 second exposure

Methods

Observations: Nights of observations typically begin prior to sunset to take calibration images. The transit data is taken in several sets in the duration of one night. Many of the dark fields were taken independent of the observing nights.

Analysis: The data is reduced and plate-solved through the CCD Data Processor in AstroImageJ. The reduced, new images are then imported into AstroImageJ and aligned using WCS (World Coordinate System). A light curve is produced via differential photometry.

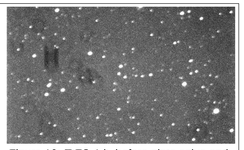


Figure 10: *TrES-1 b*, before plate solve and reduction (*7/11-7/12/23*), 30 second exposure

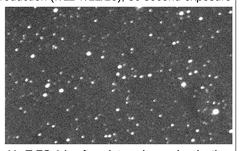


Figure 11: *TrES-1 b*, after plate solve and reduction

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